

RYUJIN -- DICE Abstract (A1 Template Mapping)

**DARPA BAA HR001126S0010 -- Decentralized Artificial Intelligence through
Controlled Emergence (DICE)**

Working draft for the A1 Abstract Template. TA1 & TA2. UNCLASSIFIED.

7-page maximum (cover sheet, ToC, bibliography excluded). 12-pt, 1" margins.

NOTE: This markdown is the content source. Final submission must be pasted into the official A1_-_DICE_Abstract_Template.docx , formatted to 12-pt with 1" margins, blue instructional text deleted, exported to PDF, and uploaded via BAAT. Fields in [BRACKETS] require your confirmation before submission.

COVERSHEET

Field	Value
Abstract Title	RYUJIN: A Doctrine-Bound, Self-Organizing Counter-Swarm Architecture for Heterogeneous Multi-Agent Autonomy
Technical Area	[X] TA1 & TA2 [] TA3
Proposer Organization	SEAN BRADLEY INC. (DBA Victory Technology Partners)
Technical POC	Sean Bradley -- [title] -- sean@victorytechpartners.com -- [phone] -- [mailing address]
Administrative POC	Sean Bradley -- sean@victorytechpartners.com -- [phone]
Other Team Members	[University sub-PI for formal analysis -- TBD; see Capabilities/Management]
Award Instrument Requested	[X] OT for Research (10 U.S.C. 4021)
Estimated Total Cost (Base + Options)	[CONFIRM -- notional working figure \$9.8M over 36 months; see Cost and Schedule]

Field	Value
Estimated Period of Performance	36 months: Phase 1 (9 mo) + Phase 2 (15 mo) + Phase 3 (12 mo)
Other solicitations this concept was submitted to	None
SAM.gov UEI	[INSERT UEI -- now issued]

1. Goals and Impact

RYUJIN delivers the **local adaptor** DICE requires: a per-agent module that makes a collective of heterogeneous AI agents coordinate without a central orchestrator (TA1) while each agent remains role-coherent and mission-aligned under adversarial pressure (TA2). The central problem DICE names -- emergence that is powerful but uncontrolled -- is exactly the failure mode we have already characterized empirically in long-horizon AI execution: **silent plan-prescription divergence**, in which individually reasonable local decisions compound into systematic drift away from mission intent. RYUJIN's thesis is that controlled emergence is achieved by **architecturally separating an inviolable doctrine layer from an adaptive coordination layer**, and coupling them through a single measurable quantity.

The motivating threat is concrete. Adversaries now field cheap, dense, leaderless drone swarms -- the "jellyfish" swarms described by front-line aircrews. A centralized orchestrator answering such a threat is a single bone whose removal collapses the skeleton. RYUJIN has no such bone: coordination is leaderless and doctrine is replicated at every node, so the collective degrades gracefully rather than catastrophically.

What is new. Most self-organizing multi-agent systems let coordination and behavior co-evolve freely, which is precisely why they accumulate entropy and lose coherence over long missions. RYUJIN inverts this. A per-agent **mission constitution** -- cryptographically signed, content-addressed, quorum-verified, and append-only after installation -- cannot be renegotiated by any peer message or local inference step. Bounded stigmergic coordination then operates **freely within the legal space the constitution defines**, while a validator gate makes

doctrine-violating collective states impossible by construction. Silent drift is detected by a conservation-and-debt-ledger mechanism and is never smoothed over; the same drift signal drives both human escalation and autonomous Byzantine reputation/slashing.

Impact if successful. A decentralized collective that (a) scales to the program's 100K-agent / 1M-interaction target with $O(n)$ interaction complexity, (b) recovers locally from agent loss or compromise without global replanning, (c) keeps heterogeneous agents doctrinally aligned across 10K-step horizons under adversarial load, and (d) surfaces infeasibility to a human operator rather than failing silently. Beyond defense, the same control kernel transfers to clinical-compliance and multi-agent enterprise systems, where silent under-delivery against contractual obligations is the dominant failure mode.

2. Technical Approach

RYUJIN is a replicated per-agent adaptor with two coupled layers over a crash-fault-tolerant compute substrate.

TA2 -- Local inference control (doctrine). Each agent carries the mission constitution: signed, content-addressed, quorum-installed role and mission invariants. Enforcement is mechanical, not prompt-based. For open-weight models, a role vector is enforced at inference time via **activation steering**; for black-box models, control is applied through context/memory engineering and mechanism design. A **role-drift detector** computes a decoherence metric (variance over total error) and a **conservation/debt ledger** tracks planned-vs-delivered obligations, surfacing infeasibility and escalating to the human responsibility boundary instead of accommodating decline.

TA1 -- Coordination and resilient fusion. Task allocation is a **bounded stigmergic auction** over a leaderless peer-to-peer bus: pheromone-biased motif selection lifts the stigmergic-flocking paradigm from robotic swarms into the cognitive/LLM layer, gated so only constitution-legal candidate actions are ever considered. Peer claims are combined by **robust context fusion** (CrossMax / Krum / trimmed-mean) with **FLTrust-style directional anchoring** that judges inputs by their direction relative to the doctrine baseline rather than by distance from peers -- defeating poisoned and adversarial inputs even when honest agents are non-IID. **Momentum-weighted reputation with slashing** and

per-node resource quotas isolate Byzantine, rogue, and kamikaze nodes; a Ben-Or randomized step breaks leaderless deadlocks.

The TA1<->TA2 coupling (our distinguishing claim). The TA2 controller publishes a **coherence horizon** -- a statistical guarantee of how many inference steps an agent remains role-coherent under current adversarial load. TA1 sets its distributed planning depth and replanning cadence to that horizon. The loop is explicit, bidirectional, and measurable; no competing architecture in scope formalizes it.

Primary attack surface, named explicitly. Mission-constitution propagation is the critical vector: a poisoned constitution means every honest node faithfully enforces the wrong rules. We mitigate in six layers: cryptographic signing with monotonic sequence numbers, content-addressed identity, quorum-gated installation (n-f attestation), append-only post-install semantics, periodic constitutional self-check, and FLTrust semantic anchoring against the operator baseline to catch adversarially crafted but validly signed doctrine.

Leveraged components (engineering substrate, not the contribution). Compute and orchestration ride on **Ray** (stateful actors, scheduling, object store; crash-fault-tolerant only -- Byzantine defense lives above it in TA1). Heterogeneous model serving uses an **OGX/vLLM** stack to present a uniform interface across different foundation models and to support fully self-hosted, air-gapped deployment for contested or enclave operation; activation-steering control reaches beneath the serving API to the vLLM layer where activation access lives. An optional DAG-BFT log (Bullshark/Mysticeti-class) is available only where a use-case requires total ordering; it is not the contribution, and framing RYUJIN as "a blockchain for agents" would be a category error.

Compute capability. RYUJIN is engineered so that scale is a function of *logical* agents, not *physical* model replicas, which keeps the compute footprint tractable. Heterogeneous agents share a small set of base-model weights served through a vLLM/OGX layer with paged-attention and continuous batching; per-agent identity is carried in lightweight context and an activation-steering role vector that adds negligible marginal cost over a shared forward pass. Because the mission constitution gates most candidate actions and the coordination layer is event-driven, the large majority of agents are idle or short-prompt at any instant, so many logical agents multiplex onto each GPU.

Coordination itself (stigmergic auction, robust fusion, reputation) is CPU/network-bound and rides the Ray substrate, not the GPUs. Concretely: Phase 1 at 500 agents / 5K interactions is feasible on a single multi-GPU node; Phase 2 at 5K agents / 50K interactions on a modest cluster; Phase 3 at 100K agents / 1M interactions via agent multiplexing plus a discrete-event simulation harness for the long tail of gated/idle agents, with elastic burst capacity for peak inference. [CONFIRM what is in hand vs. procured under the award -- e.g., current access to N x [GPU type, memory, interconnect] for Phase 1, with cloud and/or national-compute (NAIRR, DoD HPC) capacity budgeted in the cost volume for later phases.]

3. Capabilities / Management Plan

Principal Investigator: Sean Bradley (SEAN BRADLEY INC. / Victory Technology Partners). Prior work: the Duchaine System, an operational long-horizon AI control architecture that empirically characterized silent plan-prescription divergence and demonstrated conservation-law enforcement, anti-sycophancy, and infeasibility surfacing -- the direct intellectual antecedents of RYUJIN's TA2 doctrine layer. [Add: relevant engineering record; former U.S. Army Ranger -- operational doctrine/OPORD framing is first-hand, not borrowed.]

Teaming (highly encouraged at abstract stage, not yet finalized). We intend to add a sub-awardee providing formal-methods and distributed-systems depth -- specifically [PhD, graph theory / distributed consensus] -- to own the formal analysis of the coordination protocol and the Byzantine-resilience proofs. A university CS/Math department (e.g., [target institution]) is the intended academic sub. This pairs first-hand operational doctrine experience (PI) with peer-reviewed theoretical rigor (sub).

Organization and roles. PI/prime (SEAN BRADLEY INC.): architecture, TA2 doctrine/control, integration, and program management. Sub-PI: TA1 formal coordination guarantees and robust-fusion analysis. Risk is managed by phasing the highest-uncertainty work (activation-steering transfer, BFT throughput at scale) into clearly bounded milestones with black-box fallback paths.

4. Cost and Schedule

Schedule (aligned to the DICE three-phase structure):

- **Phase 1 (9 months) -- Decentralization.** Adaptor interface spec; mission-constitution module; bounded stigmergic auction v0; CrossMax fusion; drift detector + debt ledger; self-evaluation vs. a centralized baseline. Target: 500 agents / 5K interactions.
- **Phase 2 (15 months) -- Adversarial robustness.** BFT context fusion + reputation/slashing; 1K-step adversarial coherence; activation steering for open-weight models; scale to 5K agents / 50K interactions; team-vs-team eval.
- **Phase 3 (12 months) -- Scaling.** $O(n)$ interaction complexity; 10K-step adversarial coherence; 100K agents / 1M interactions; final leaderboard and team-vs-team.

Cost (notional -- **[CONFIRM all figures]**): Estimated total
[~\$9.8M] over 36 months (Base + Options), broken down by phase with **compute cost itemized separately** per the template:

Phase	Labor + sub	Compute (itemized)	Phase total
Phase 1 (9 mo)	[\$1.6M]	[\$0.5M]	[\$2.1M]
Phase 2 (15 mo)	[\$3.3M]	[\$1.4M]	[\$4.7M]
Phase 3 (12 mo)	[\$2.0M]	[\$1.0M]	[\$3.0M]
Total	[\$6.9M]	[\$2.9M]	[\$9.8M]

[These are placeholders to make the structure explicit. Right-size against the realistic two-person-plus-sub labor base, your actual compute plan, and efficient-inference assumptions the template asks you to consider. Compute is deliberately broken out because the BAA requires it.]

5. Publications (up to 10, 2016+; team-member names highlighted)

1. **Bradley, S.** The Duchaine System: constraint-satisfaction and conservation-law enforcement for long-horizon AI execution (2026). [link]
2. [Add up to nine additional relevant references; where the sub-PI is added,

highlight their peer-reviewed distributed-systems / graph-theory papers here.]

6. Bibliography (does not count against page limit)

Reynolds (1986) Boids; Grasse (1959), Dorigo (1992) stigmergy/ACO; Castro (1999) PBFT; Yin et al. (2019) HotStuff; Spiegelman et al. (2022) Bullshark; Mysten Labs (2023) Mysticieti; Blanchard et al. (2017) Krum/Bulyan; Cao et al. (2020) FLTrust; Ben-Or (1983) randomized consensus; Zou et al. (2023), Turner et al. (2023) activation steering; Moritz et al. (RISELab) Ray; Tomasev et al. (2025) distributional AGI safety; Simon (1956) bounded rationality. [Format full citations with links in the final document.]

Open items before submission (your inputs)

1. ☐ UEI string -> cover sheet
2. ☐ Technical/Admin POC contact block
3. ☐ Confirm award instrument (recommended: OT for Research)
4. ☐ Confirm/replace all cost figures and the compute plan
5. ☐ Compute-capability paragraph (what hardware is in hand vs. to be procured)
6. ☐ PI bio specifics; intended sub-PI and university to name
7. ☐ Publications list (finalize; add sub-PI papers once partner is confirmed)
8. ☐ BAAT account created for upload